

Yukta Kasina

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PROFESSIONAL SUMMARY

ML Engineer with a Master's in AI and real-world deployment experience across NLP, CV, and RL systems. Passionate about closing the gap between research and production, building models that don't just work in notebooks.

EDUCATION

Northeastern University, Boston, MA

Sep 2024 – May 2026

Master of Science in Artificial Intelligence (GPA: 3.625)

SRM Institute of Science and Technology, India

Sep 2020 – Jun 2024

Bachelor of Technology in Computer Science and Engineering (GPA: 3.74)

SKILLS & TOOLS

Languages: Python, C++, Java, SQL, TypeScript

ML/DL: LLM Fine-tuning & Pretraining, Recommendation Systems, NLP, Deep Learning, Zero-shot Learning, Transformer Architectures

Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, Scikit-learn

Data: Data Gathering & Preprocessing, Pandas, NumPy, Matplotlib, Seaborn, Apache Spark

MLOps: Model Deployment, CI/CD (GitHub Actions), Pipeline Automation, Docker, Experiment Tracking, Model Monitoring, Data Validation

Systems: Scalable Distributed Systems, REST APIs, Microservices, System Design

Cloud: AWS (SageMaker, Lambda, S3), GCP, Cloud Deployment Workflows

PROFESSIONAL EXPERIENCE

Kirusa, Inc.

Aug 2023 – Nov 2023

Machine Learning Intern

- Built and deployed NLP models using HuggingFace Transformers, creating end-to-end pipelines and inference APIs.
- Automated data validation and preprocessing, improving pipeline reliability.
- Implemented monitoring and evaluation workflows, increasing production accuracy by 12%.

AICTE

May 2023 – Jul 2023

Machine Learning Intern

- Built deep learning-based recommendation components for image classification workflows.
- Developed model evaluation frameworks to assess performance against domain-specific benchmarks.
- Produced visual and written reports for both technical and non-technical stakeholders.

PROJECTS

Explainable AI for Tumor Classification — *ResNet, PyTorch, Grad-CAM, SHAP*

- Built ResNet/VGG models on preprocessed tumor image datasets to classify benign vs. malignant growths.
- Applied Grad-CAM and SHAP to visualize learned features, aiding model transparency and trust.
- Enhanced model generalization using data augmentation and 5-fold cross-validation techniques.

Picasso: RL-Based Painting Agent — *TD3, PyTorch, CelebA, LearningToPaint*

- Replaced DDPG with TD3 in the LearningToPaint framework, improving training stability on CelebA portrait generation.
- Trained agent to 500k steps on an RTX 4060, achieving mean distance 0.026 against target images.
- Extended reward shaping and action space tuning to produce coherent brushstroke-level reconstructions.

Sentilyrics: Lyrics Mood Classifier — *NLP, PyTorch, Transformers, BERTs*

- Built an end-to-end NLP pipeline to classify song lyrics into mood categories using large-scale datasets (5M+ samples).
- Fine-tuned Transformer-based models (BERT) alongside LSTM/CNN baselines, achieving 87% accuracy.
- Designed modular preprocessing and training workflows, enabling scalable experimentation and reproducibility.